FlytBase UAV Systems Engineer Assessment

Multi-Drone Coordination System

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Document Title	Validation and Testing Plan
Version	1.0 — Initial Submission
Test Phases	5 phases: Simulation Single Pair Row Full Grid Adversarial
Total Test Scenarios	6 named scenarios + stress testing + adversarial injection
Safety Standard	Minimum Separation Distance (MSD) > 8m at all times

1.Test Philosophy and Risk Model

The validation approach follows a **progressive risk model**: every phase must pass all acceptance criteria before escalating to the next. This prevents live-drone testing of scenarios that simulations reveal unsafe and ensures each layer of complexity is independently validated before combination.

Phase	Environment	Max Drones	Risk Level	Objective
Phase 0	Simulation (SITL)	16 virtual	None	All SOPs digitally validated; no physical risk
Phase 1	Single Drone Live	1	Very Low	Baseline drone behaviour confirmed
Phase 2	Pair Testing	2	Low	Same-column and cross-row scenarios
Phase 3	Row Testing	4–8	Medium	Full row simultaneous operations
Phase 4	Full Grid	16	Medium-High	All docks operating simultaneously
Phase 5	Adversarial	8–16	High (controlled)	Deliberate failure injection; stress limits

Table 1 — Six-phase progressive test plan with escalating drone counts and risk levels.

GATE PRINCIPLE	No test phase begins until the previous phase achieves a 100% pass rate across all scenarios. Any failure at Phase 4 or 5 triggers a mandatory return to Phase 0 (simulation) for root cause analysis before live testing resumes.

2.Rule testing scenarios

Six named test scenarios cover the full SOP ruleset. Each scenario is run first in simulation (Phase 0), then replicated live. Additional stress variants are noted where applicable.

T-01	**Vertical Corridor Exclusivity**
OBJECTIVE	Verify that only one drone per column can be in the 0–15m corridor simultaneously.
SETUP	D1 and D5 (both Column 1) simultaneously submit takeoff requests. Both are immediately authorized by default to test the system gate.
EXPECTED	MDCS approves D1. D5 is held with 'Corridor Occupied — ETA X second message. D5 motors remain disarmed until D1 altitude > 16m.
PASS IF	Minimum separation > 8m always maintained. D5 does not begin ascent until D1 clears 16m AGL. Corridor lock releases automatically.
FAIL IF	Both drones simultaneously in 0–15m band with < 8m separation. D5 begins ascent before MDCS clearance.

T-02	**Row Altitude Layer Separation**
OBJECTIVE	Verify drones from different rows to maintain vertical separation during simultaneous horizontal transit over the same ground area.
SETUP	D2 (Row 1, 40m) and D6 (Row 2, 35m) are given targets on opposite sides of the grid, ensuring their horizontal paths overlap for 10 seconds.
EXPECTED	Drones transit at their respective altitudes. 5m vertical separation maintained throughout. No lateral avoidance manoeuvre is needed.
PASS IF	Vertical separation 4m always during transit overlap. Neither drone receives any conflict alert.
✗ **FAIL IF**	Vertical separation < 3m at any point. Any drone deviates from the assigned altitude layer.

T-03	**Priority Override — Low Battery**
OBJECTIVE	Verify P2 (low battery) priority correctly preempts P4 (routine) corridor access.
SETUP	D9 (battery 22%, P4 routine mission) is in Column 1 corridor descending. D5 (battery 11%, P2 urgent) simultaneously requests the same corridor.
EXPECTED	MDCS immediately reassigned P2 to D5. D9 is commanded to hold at 16m in hover. D5 gets corridor clearance. D9 resumes after D5 lands.
PASS IF	D5 lands within 4 minutes of declaring P2 status. D9 holds without descending below 15m until corridor clears.
✗ **FAIL IF**	D5 queued behind D9 despite P2 status. Any drone-drone separation < 8m in corridor.

T-04	**Simultaneous 8-Drone RTH**
OBJECTIVE	Verify that all 8 drones from two rows simultaneously executing RTH produces no conflicts.
SETUP	8 drones (D1–D8, Rows 1 and 2) are all on active missions. Trigger E-STOP simultaneously for all 8.
EXPECTED	Each drone navigates to its column corridor at its row altitude. Column corridors are sequenced automatically. All 8 land successfully.
✓ **PASS IF**	All 8 drones land within 12 minutes. No drone-drone separation < 8m recorded anywhere in telemetry logs.
✗ **FAIL IF**	Any drone-drone separation < 5m. Any crash, flyaway, or non-responsive drone. Any drone landing on wrong dock.

T-05	**MDCS Software Failure — Autonomous RTH**
OBJECTIVE	Verify all drones safely RTH without MDCS guidance when MDCS fails mid-operation.
SETUP	6 drones are on active missions at different altitudes and positions. MDCS software is forcibly terminated in mid-operation.
EXPECTED	All drones detect MDCS loss within 5 seconds. Each initiate autonomous RTH with (dock_number × 5s) stagger delay using onboard GPS.
✓ **PASS IF**	All 6 drones land within 10 minutes. No drone-drone separation < 5m. Backup console shows drone positions throughout.
✗ **FAIL IF**	Any crash, lost drone, or drone-drone separation < 5m. Any drone failing to initiate autonomous RTH within 10 seconds of MDCS loss.

T-06	**Wind Stress — Simultaneous Dual Drone**
OBJECTIVE	Verify that SOPs remain effective, and altitude layers are maintained during 10 m/s crosswind.
SETUP	Two drones from different rows (D3 and D7) fly intersecting ground tracks on a day with a confirmed 10 m/s crosswind at transit altitude.
EXPECTED	Drones apply heading corrections to maintain track. Altitude layers maintained ±2m. Mission completed successfully.
PASS IF	Both drones are complete mission. Altitude maintained within ±3m of assigned layer. Separation > 8m maintained.
✗ **FAIL IF**	Any drone drifting > 5m from assigned altitude. Any drone-drone separation < 8m. Mission abort due to the wind.

3.Safety Validation Methods

Telemetry-Based Separation Analysis

Every flight log GPS position, altitude, and velocity at 10 Hz. An automated post-flight script calculates the Minimum Separation Distance (MSD) between every drone pair at every timestep. Results are visualised as:

- Separation distance time-series graph shows MSD over flight duration with 8m and 12m threshold lines marked.
- Closest approach heat map shows where in the airspace the smallest separations occurred, flagging hotspot zones.
- Safety margin score: MSD / 8m — target > 1.5 on all flights. Score < 1.0 is an automatic violation.
- Histogram of all pairwise separations: distribution should be right shifted, with no observations below 8m.

Fault Tree Analysis (FTA)

A formal FTA document maps all top-level hazard events to contributing causes. Each SOP rule maps to one or more FTA mitigations. The FTA is updated whenever a test gap is identified.

Hazard Event	Target Probability	Mitigation SOPs
Two drones collide in corridor	< 1×10■■ per flight hour	SOP-TKO-1, SOP-LND-1 Column corridor locking
Drone crash due to battery exhaustion	< 1×10■■ per flight hour	SOP-PRI P2 Pre-flight battery > 30% gate
Comms-lost drone re-enters occupied corridor	< 1×10■■ per event	SOP-EMG-E1 20m avoidance buffer rule
MDCS failure causes uncoordinated RTH	< 1×10■■ per year	SOP-EMG-E4 Staggered autonomous RTH
Wind drift causes altitude layer violation	< 1×10■■ per flight hour	Wind sensor alerts ±2m altitude tolerance gate

Acceptable vs. Unacceptable Outcomes

MSD consistently > 12m	Excellent — green zone	Continue current phase
MSD between 8m and 12m	Acceptable — yellow zone	Flag for review; monitor trend
MSD between 5m and 8m (near-miss)	Unacceptable — red zone	Ground all drones; root cause analysis before resuming
MSD < 5m (critical near-miss)	Critical failure	Full operational pause; mandatory SOP revision; return to Phase 0
Physical collision	Catastrophic failure	Full stand-down; formal incident investigation; regulatory notification
MDCS failure with safe autonomous RTH	Acceptable (designed behaviour)	Log; review RTH timing; no suspension required
MDCS failure + any drone crash	Critical failure	Full operational pause; MDCS architecture review

4.Efficiancy Measurement

Throughput Measurement Protocol

Run a 2-hour standardised test scenario with each drone assigned a fixed target 200m away. Count completed round-trip missions per hour. Compare against theoretical maximum (if there is zero wait time). Efficiency = Actual throughput / Theoretical max × 100%.

Scenario	Theoretical Max (missions/hr)	Target Actual	Acceptable Min	Efficiency Target	Primary Bottleneck
1 drone (baseline)	12	12	11	100%	—
4 drones (1 per row)	48	44	40	91%	Column corridor queuing
8 drones (2 per row)	72	60	52	83%	Same-column landing stagger
12 drones (3 per row)	96	75	65	78%	Corridor saturation in peak rows
16 drones (all docks)	120	90	75	75%	Simultaneous RTH stagger delay

Table 2 — Throughput targets by operational scale. Values are conservative estimates.

Bottleneck Identification

Column Corridor Contention

Most likely bottlenecks when 4 drones in the same column all need to land within a short window; each must wait for the corridor to clear. Measured by per column queue waits time logs.

Mitigation: Mitigation under test: stagger mission launch times for same-column drones by 3–5 minutes to prevent simultaneous RTH clustering.

Priority Inversions

P4 drones waiting excessively for P1/P2 drones to clear. Measured by: max queue wait time per drone per session.

Mitigation: Mitigation: maximum P4 wait cap of 5 minutes; if exceeded, drone is offered alternative mission or standby mode.

Weather-Induced Speed Reduction

Strong winds force drones to reduce to 3 m/s over the grid, reducing mission cycle time. Measured by: avg mission duration vs. baseline on zero-wind days.

Mitigation: Mitigation: dynamic wind-based speed limits communicated from MDCS; throughput KPI adjusted for weather conditions.

Comparison: Theoretical vs. Practical Throughput

The gap between theoretical and practical throughput stems entirely from coordination overhead (corridor wait times, priority staggering, weather reductions). The MDCS targets reducing this gap through intelligent scheduling. Target: practical throughput 75% theoretical at full 16-drone scale. Any scenario achieving < 65% triggers a scheduling algorithm review.

5.Iterative Improvement Process

The iterative improvement cycle converts test findings into systematic SOP enhancements. The process is executed after every test session and formally reviewed monthly.

STEP 1 TELEMETRY REVIEW	Automated post-flight scripts analyse all telemetry logs. Flag any MSD < 12m (early warning) or < 8m (violation) events. Generate separation distance histograms.
STEP 2 ROOT CAUSE ANALYSIS	For every flagged event: identify which SOP rule was absent, ambiguous, or violated. Map to Fault Tree Analysis (FTA) document to identify systemic gaps.
STEP 3 SOP AMENDMEN T	Draft specific SOP amendment with version number and change rationale. Peer review by minimum of 2 engineers before adoption. Document impact on existing rules.
STEP 4 REGRESSION TEST	Re-run the exact test scenario that exposed the gap in SITL first, then live. Confirm amendment resolves the gap without creating new issues in adjacent scenarios.
STEP 5 DOCUMEN T UPDATE	Update all 3 assessment documents with new version number. Maintain full change log. Re-distribute to all operators. Brief team on what changed and why.

Fig. 1 — Five-step iterative improvement cycle applied after every test session.

Response Tiers by Severity

Trigger	Response	Timeline
MSD < 12m (early warning)	SOP clarification or minor rule addition	Within 1 week; no operations suspension
MSD < 8m (violation)	Mandatory SOP amendment + full regression test	Before next live flight; SITL test first
Any physical crash or loss	Full operational pause + formal root cause analysis	Minimum 1-week stand-down; regulatory notification
3+ same-category incidents	Architectural review — consider system redesign	2-week analysis period; external safety audit recommended

Incremental vs. Major Redesign Decision Criteria

The default preference is incremental SOP refinement, which can be tested and deployed quickly without disrupting the overall system architecture. A major redesign is triggered only when:

- Three or more failures point to the same fundamental architectural flaw (e.g., the altitude layer system is shown to be insufficient for the target density).

- A new operational requirement (e.g., adding a 5th row, or BVLOS authorization) is incompatible with current SOP structure.
- An independent safety audit recommends architectural changes.
- Regulatory changes require fundamental rethinking of airspace structure.

DESIGN PRINCIPLE	Prefer incremental over major redesign wherever possible. Every major redesign resets validation to Phase 0 (simulation) and requires full regression testing across all 6 scenarios before live operations resume. The cost of a major redesign is 4–8 weeks of testing.	

Continuous Improvement KPIs

Continuous Improvement KPI	Target Trend	What It Indicates
SOP Version Increment Rate	1–2 per month during active ops	Indicates active learning and refinement
Mean Time Between Incidents (MTBI)	Increasing trend month-over-month	Core safety improvement indicator
Average Safety Margin (MSD / 8m)	Increasing trend; target > 2.0 at maturity	Indicates growing separation buffers
Throughput Efficiency	Increasing trend toward 85% at full scale	Indicates scheduling improvements
Operator Response Time to P1 Alert	Decreasing trend; target < 3 seconds	Indicates operator training effectiveness

6.All Test Summary

ID	Scenario	Phases	SOPs Tested	Pass Criteria Summary
T-01	Vertical Corridor Exclusivity	P0–P2	SOP-TKO-1	MSD > 8m; corridor sequencing correct
T-02	Row Altitude Separation	P0–P4	SOP-1.1, SOP-TRN-1	Vertical sep □ 4m during transit overlap
T-03	Priority Override — Battery	P0–P3	SOP-PRI P2, SOP-LND-1	P2 drone lands within 4min of declaration
T-04	8-Drone Simultaneous RTH	P0–P4	SOP-LND, SOP-PRI	All 8 land < 12min; no MSD < 8m
T-05	MDCS Failure Autonomous RTH	P0–P4	SOP-EMG-E4	All land < 10min; no MSD < 5m
T06	Wind Stress — Dual Drone	P0–P4	SOP-TRN-1, SOP-1.1	Alt maintained ±3m; MSD > 8m